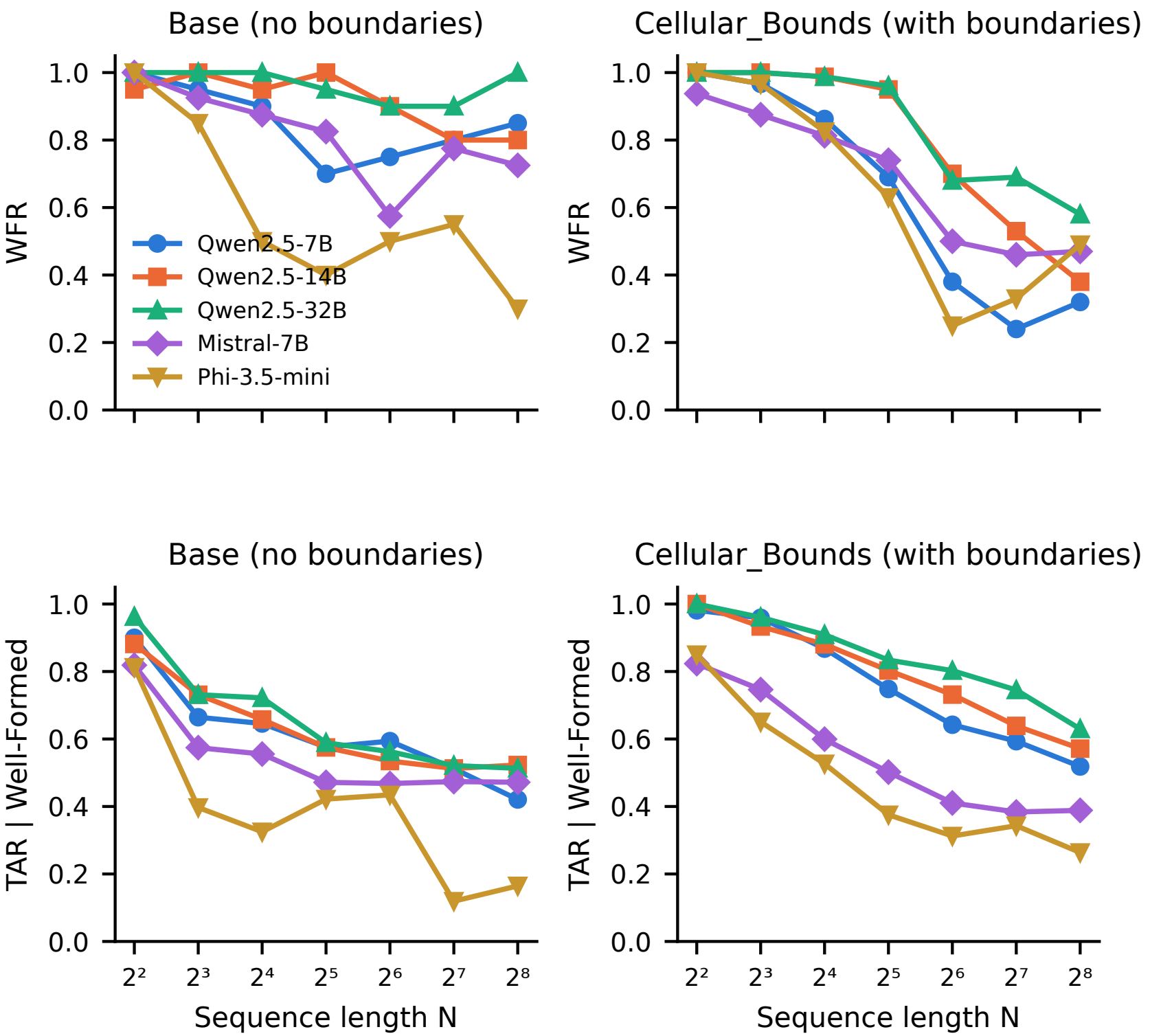cyclic_shift -- failure-mode diagnostics by model
N on x-axis (not pooled), alphabet=2